

An old AI-memory backup can look current.

After a crash, the agent operator restores a private snapshot. A separate controller needs evidence before deciding whether the agent may resume.

Verify the history,
not the memory.

THE SILENT ROLLBACK

One restore. Two different histories.

01

Snapshot 1

Private memory checkpoint.

02

Snapshot 2

Newer state is committed.

03

Snapshot 3

Canonical committed head.

04

Restore 1

Operator tries to continue from the old root.

The local database can reopen. An independent reviewer still needs to know whether that state may extend the shared history.

WHY WEB3 BELONGS HERE

The runtime operator cannot be the only keeper of the checkpoint.

A shared registry records the current predecessor and configured authority. Raw memory stays off-chain.

PRIVATE RUNTIME

Stores the snapshot

The operator can restore or replace local files after a crash or migration.

REGISTRY AUTHORITY

Controls succession

The Solidity registry checks sequence, predecessor root, and authorization.

INDEPENDENT REVIEWER

Replays the evidence

A separate Rust verifier checks the portable history without trusting the website.

From a private snapshot to a replayable decision.

01

SQLite

Three synthetic private snapshots.

02

Rust

Versioned commitments and evidence.

03

Solidity

Ordered state and authority rules.

04

Rust/revm

Execute the published bytecode locally.

05

Inspector + CLI

Explain and independently replay the result.

Stack: Solidity 0.8.36 · Rust 1.97.1 · Dioxus/WASM · SQLite · revm · Ethereum Sepolia for a separate public observation

FALSIFIABLE RESULT

Transition 4 names the old root. The registry rejects it.

Canonical head: transition 3. Attempted predecessor: actual root after transition 1.

REJECTED BAD_PREVIOUS_STATE

snapshot 1 → KNOWN_HISTORICAL_CHECKPOINT
reason: current head is snapshot 3
REHEARSE_ONLY → HELD
reference loader not invoked

The contract check rejects the stale root. Separately, the framework-neutral local reference runtime holds snapshot 1 before its loader. Production integration is not claimed.

M. MEMORYLINEAGE
Inspect
History
Tampering Lab
Verify
GITHUB / PUBLIC SOURCE

TAMPERING LAB / FALSIFICATION WORKSPACE

PUBLISHED LOCAL REVN / RUST-WASM VERIFIED / DEMO-SPACE-V2-SILENT-ROLLBACK

Tampering Lab

Restore an older snapshot, then try to continue the newer history. The result explains exactly what the registry accepts or rejects.

01 Restore

02 Compare

03 Replay

04 Verdict

CHOOSE A TEST

LOCAL EVIDENCE

Silent Rollback

restore a stale private snapshot and try to continue

CORPUS

Sequence Gap

skip the next canonical sequence number

CORPUS

Parallel History

present another successor from the same predecessor

CORPUS

Wrong EOA Signer

reuse the transition with an unrelated signer

CORPUS

Locator Substitution

change a locator commitment after signing

CORPUS

Wrong Chain Domain

reuse a signature from another chain or contract domain

WHAT WE ARE COMPARING

LOCAL EVIDENCE: REJECTED

Silent Rollback

SYNTHETIC FIXTURE / CANONICAL

SNAPSHOT

SNAPSHOT 1

Language preference recorded

SNAPSHOT 2

Evidence-backed claim policy recorded

SNAPSHOT 3 / COMMITTED

Raw-memory privacy boundary recorded

protocol root 0xa80ff09703...0666f2f0

LOCAL RESTORE ATTEMPT

RESTORE SNAPSHOT 1

Language preference recorded

ATTEMPT TRANSITION 4

stale predecessor 0x3635645aa8...d14f0ee0

Run Silent Rollback →

Runs published local evidence; no transaction is sent.

LOCAL EVIDENCE: REJECTED

×

LOCAL EVIDENCE: REJECTED

PUBLISHED LOCAL REVN / BAD_PREVIOUS_STATE / attempted sequence 4 / stale root 0x3635645aa8...d14f0ee0 from transition 1

The published Rust/revm run rejected transition 4.

The browser has independently replayed the evidence: the stale predecessor is transition 1's root, while the canonical local head is transition 3. Run 'cargo xtask verify' to regenerate the SQLite fixture and execute the Solidity bytecode again.

Reset rehearsal

Probe separate Sepolia

WHAT THIS PROVES

Stale predecessor rejected

The attempted root is not the canonical head.

Canonical succession enforced

Sequence and parent state are checked together.

Raw memory stays private

Only commitments and metadata enter the evidence.

SCREENSHOT FROM THE LOCAL STATIC INSPECTOR · DEMO SPACE V2

05 / 08

DO NOT TRUST THE DASHBOARD

Change one commitment. Replay must fail.

Export the same evidence bundle from the Inspector and check it with the independent Rust CLI.

```
original → BUNDLE REPLAY VERIFIED
tampered → TRANSITION_ID_MISMATCH
restored → BUNDLE REPLAY VERIFIED
```

Replay confirms bundle consistency, not canonical registry provenance. Receipts bind the decision to the supplied bundle and policy.

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Detailed checks performed on the loaded evidence bundle.

Schema	PASS
Registry address	ADDRESS FORMAT ONLY
Transition IDs	PASS
State roots	PASS
Sequence continuity	PASS
Predecessor continuity	PASS
Authority timeline	TIMELINE ROUND
Transition authorization proof	EOA SIGNATURES VERIFIED
Privacy boundary	PASS
Attack evidence	PASS

The browser changes the first locatorCommitment in a copy of the loaded evidence. A valid bundle must fail closed before it can be trusted.

ORIGINAL

locatorCommitment / 0x16...

→

TAMPERED COPY

locatorCommitment / 0xff...

Tamper one field
Restore original

Original evidence restored; local replay passed.

BROWSER RESULT

✓ BUNDLE REPLAY VERIFIED
The original bundle passes the Rust/WASM replay checks.

The browser verifier recomputes the bundle. A changed commitment cannot silently preserve the original transition identity.

Open Public Evidence →

RECOVERY DECISION RECEIPT LOCAL RECEIPT VERIFIED

Can another process verify the resume decision?

A Recovery Decision Receipt binds one private snapshot commitment to the head reconstructed from the supplied bundle and records the decision. It contains no raw memory and does not claim to enforce a production runtime.

DECISION READOUT

✓ RECEIPT VERIFIED
Local receipt replay: CURRENT_HEAD / RESUME_ALLOWED / decision 0xeba538142f46...f1448e74.

PORTABLE ARTIFACT

Export the decision, then falsify it.

The Rust CLI can verify the same receipt outside the browser against the same V2 bundle.

Recovery preflight around pinned registry semantics.

Built and evidenced here

- Current, historical, divergent, and unverified classification.
- Policy-bound recovery receipts and protected local loader.
- Independent replay, authority timeline, and tamper evidence.
- Rust/WASM Inspector and a reproducible local incident.

Outside the current proof

- Semantic truth or safety of the private memory.
- Production agent adoption or off-chain availability.
- Public deployment of this exact Demo Space V2 history.
- Canonical bundle provenance, complete history recovery, historical ERC-1271 replay, or RPC consensus.
- Authorship of ERC-8350, EIP-712, or ERC-1271.

Make recovery reviewable before an agent resumes.

For managed agent systems with separate operators, controllers, and auditors, MemoryLineage offers a narrow shared answer to an otherwise private restore decision.

Try the incident

Open the Inspector, check an old restore, and inspect the exact rejection.

Replay the evidence

Run the submission CLI verifier from the named source revision.

Extend the proof

Collect independent human reproduction and a public same-space deployment before making those claims.

memorylineage.pages.dev · github.com/AndroLay/MemoryLineage